

ORAL CAVITY IN DISEASE

Dr. Syed Mukith

Chapter – 1

DEVELOPMENTAL DISEASES OF ORAL AND PARA ORAL STRUCTURES

Dr. Syed Mukith

Learning Outcomes of Chapter

Developmental Diseases affecting Oral and Para-Oral Structures

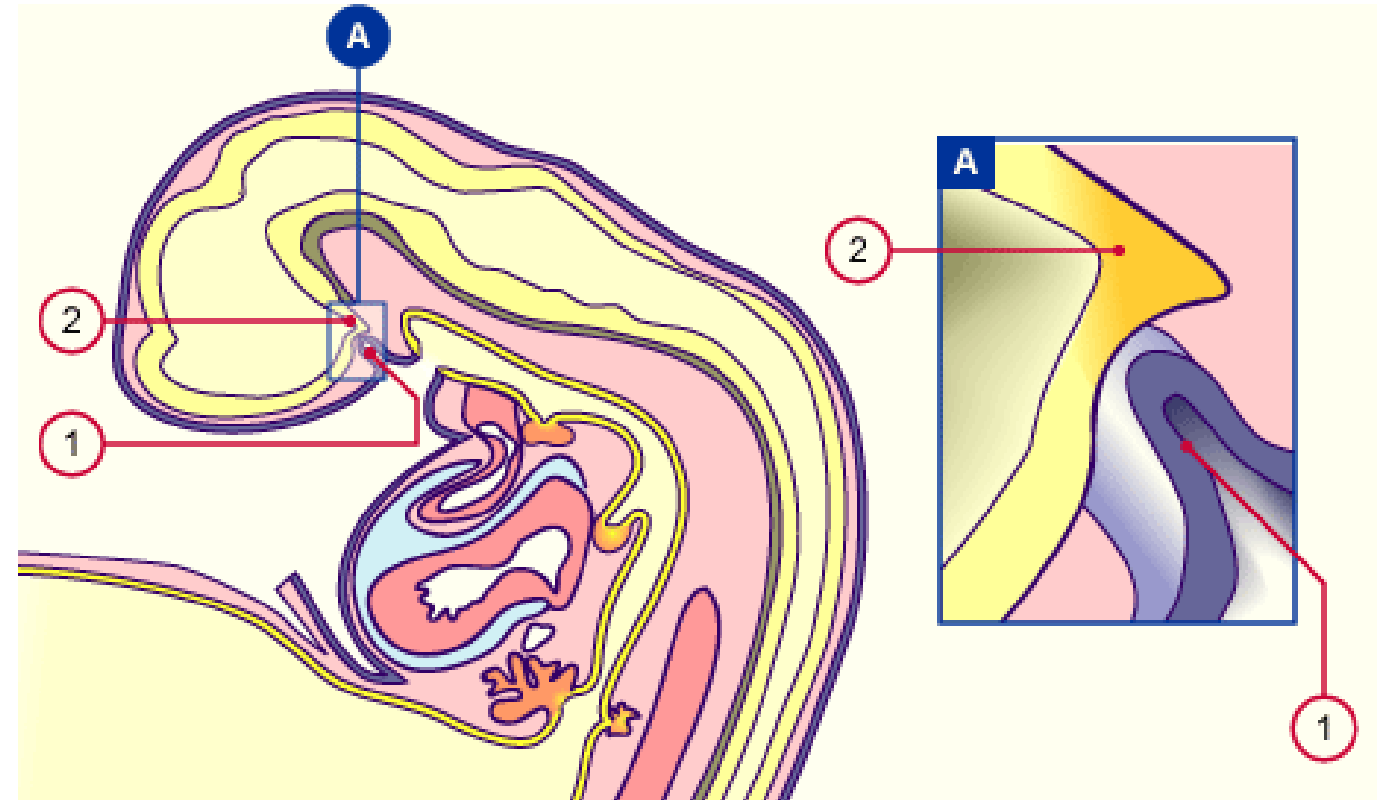
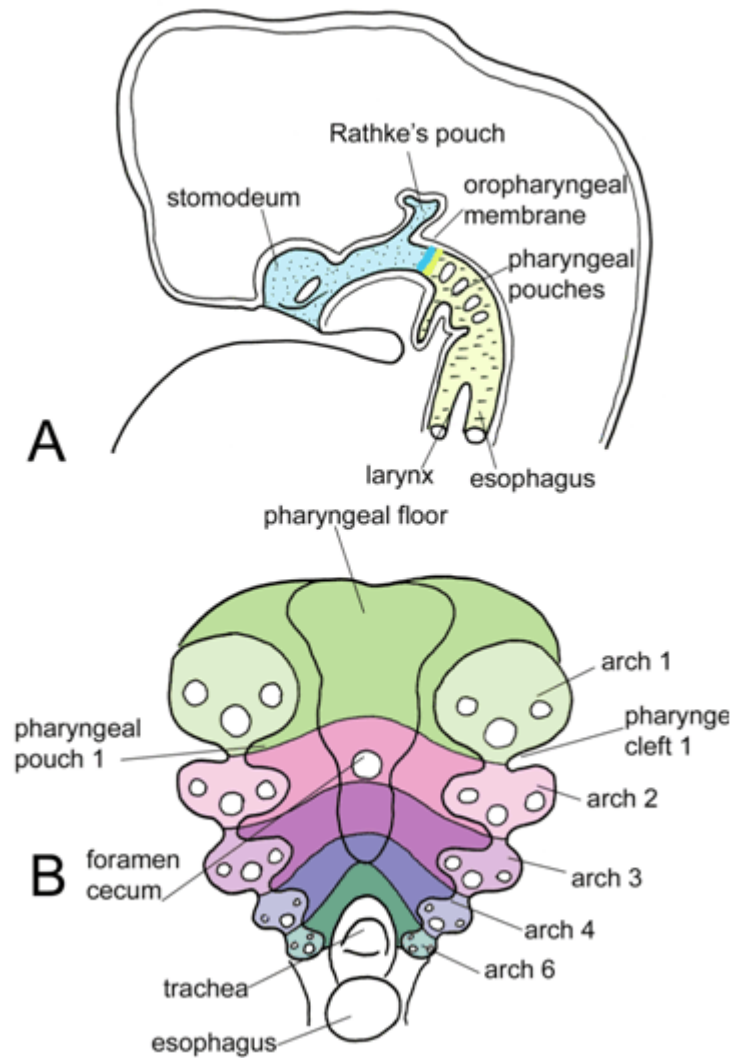
CLO	LO	By the end of this chapter, students should be able to:	
1.1 2.2 2.3	1	<u>Memorize the classifications of</u> the developmental diseases affecting oral and paraoral structures.	
	2	<u>Describe the pathophysiology and criteria for diagnosis</u> of developmental diseases affecting hard and soft tissues of oral cavity.	
	3	<u>Interpret clinical and radiographic features</u> of developmental diseases affecting number, structure, shape and size of teeth.	
	4	<u>Explain the clinicopathologic features</u> of developmental diseases affecting soft tissues and bone.	

Essential reading:

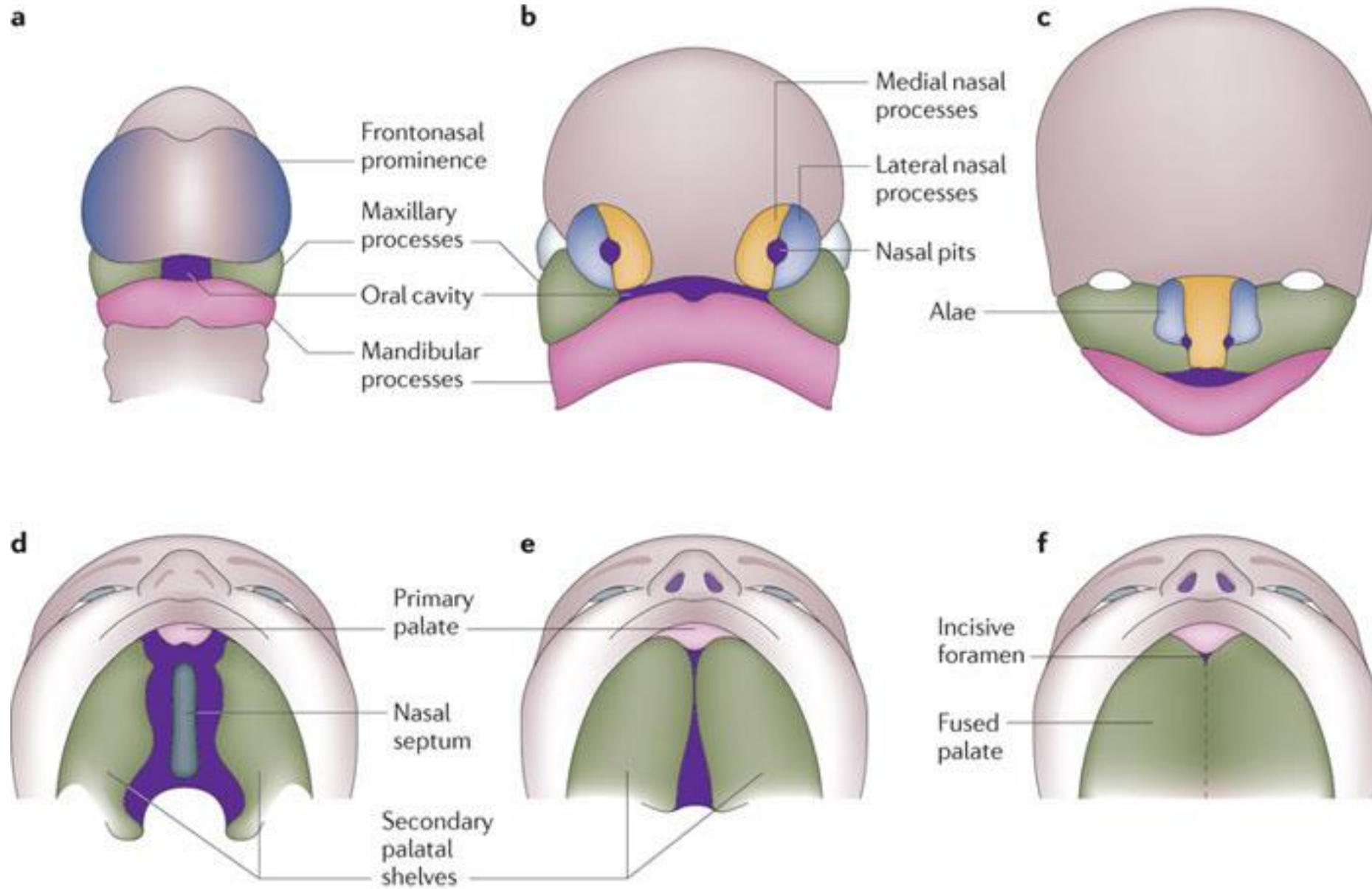
- Regezi & Sciubba: Oral Pathology Clinicopathologic Correlation; 2017 Elsevier (7th edn),
Chapter 15 - Metabolic and Genetic Diseases
Chapter 16 – Abnormalities of teeth
- Electronic Link: Tongue Diseases (Available in E-learning)

Overview of Development of Face, Mouth and Teeth

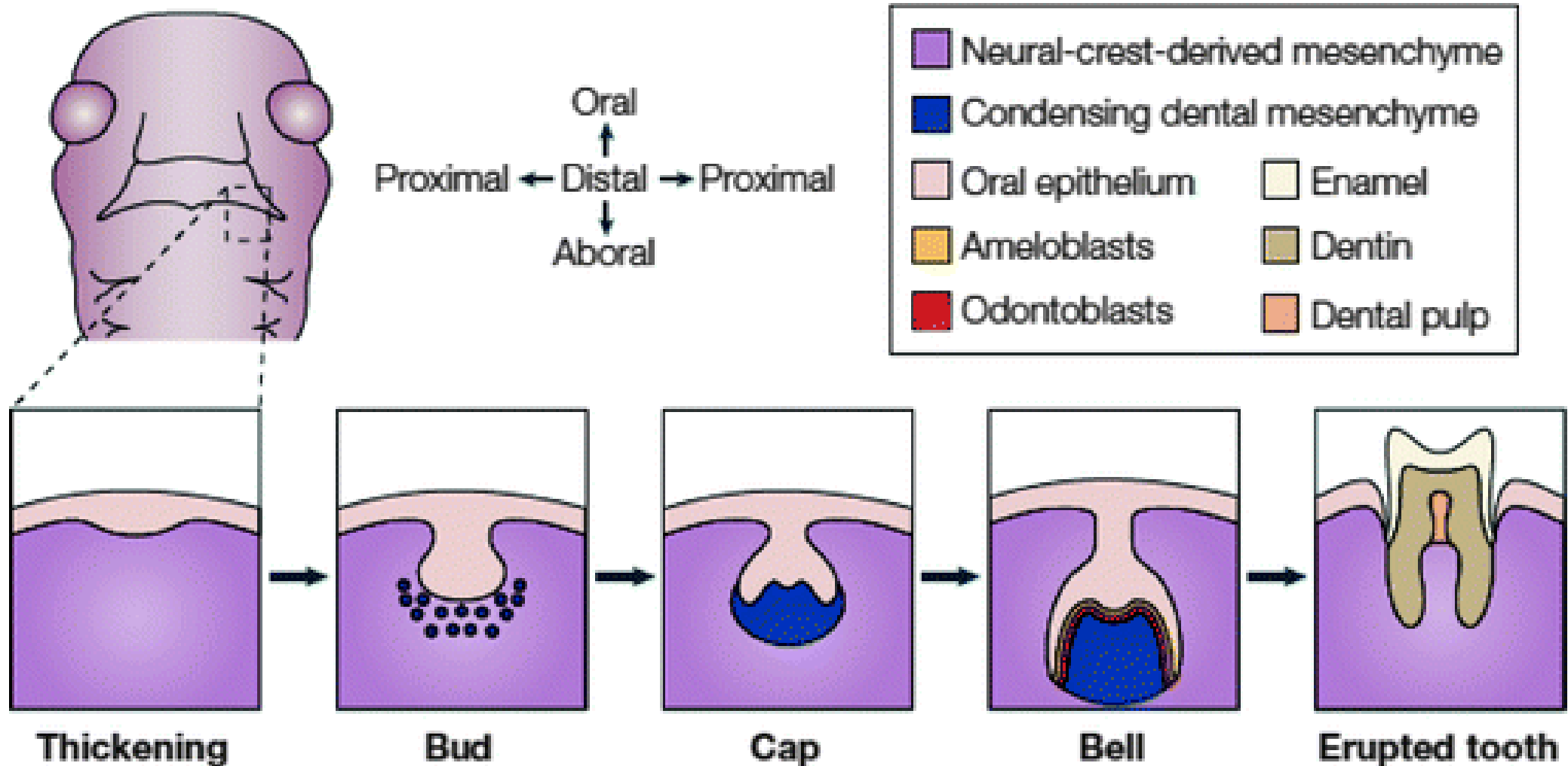
Embryonic Development of Oral and Paraoral Structures



Face and Palate Development



Tooth Development Stages



Video Links for Development of Face and Teeth

Development of Face

<https://www.youtube.com/watch?v=zNgG8WmRXil>

Development of Teeth

<https://www.youtube.com/watch?v=t3hR2YGdqWk>

Pathophysiology of Developmental Diseases

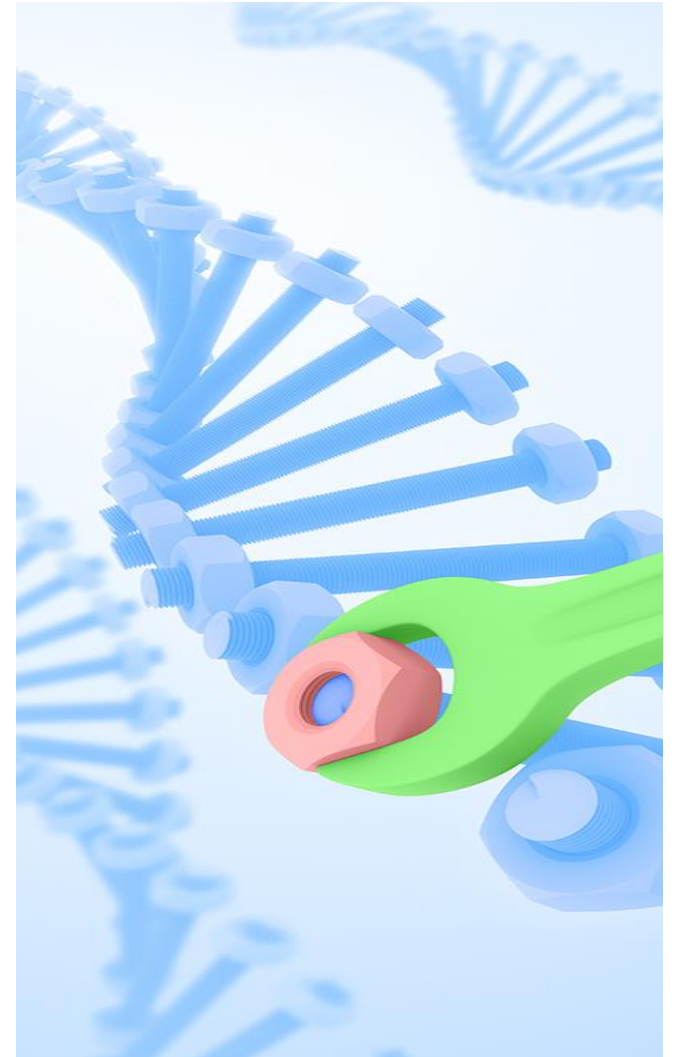
Terminologies

- **Congenital Meaning** existing at or dating from birth
- **Developmental Disorder:** a condition that is typically marked by delayed development or impaired function especially in function, communication, learning, cognition, behavior, socialization.
- **Genetic Disorder** is a condition that occurs as a result of a mutation to DNA.

Causes of Developmental Disturbances

A. Mutagenesis

1. Infections - Rubella/Influenza
2. Physical Injuries Pressure Temperature changes
Radiation
3. Hormones ... Diabetes mellitus Hyperthyroidism
Hypothyroidism ACTH Cortisone Androgenes Estrogen
4. Nutrition..... Deficiencies of – Vitamin A, B Complex, D, E,
K, Proteins, PUFA, Potassium.
5. Respiration... Hypoxia, Carbon dioxide excess, Carbon
monoxide Anesthesia with ether gas-oxygen.



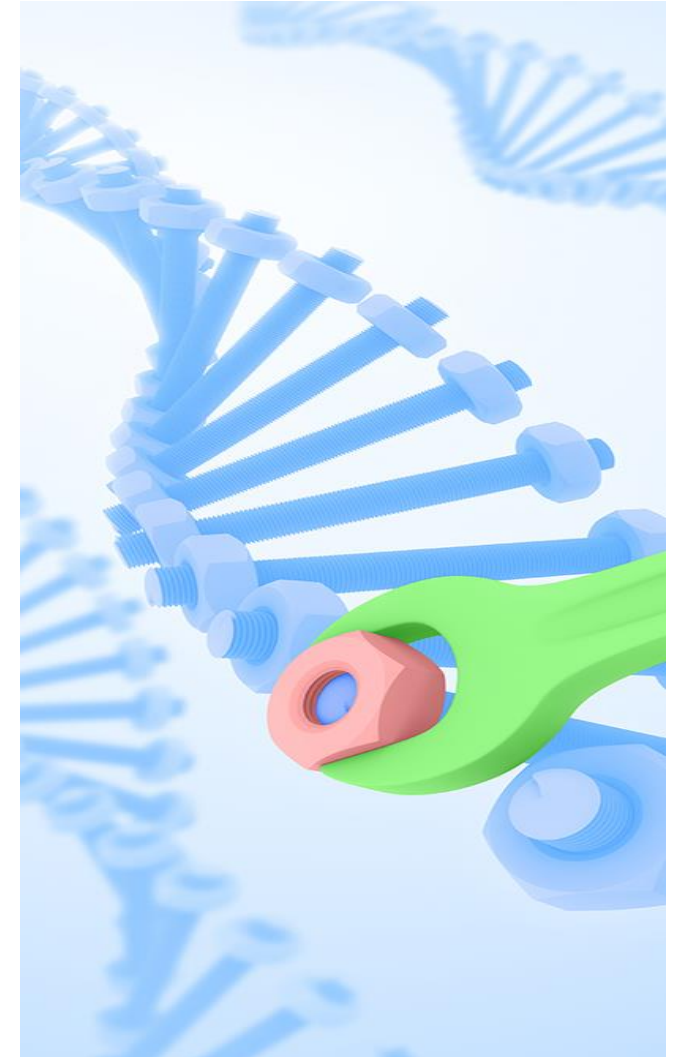
Causes of Developmental Disturbances

A. Mutagenesis

6. Miscellaneous drugs and chemicals ... Nicotine, Sulfonamides, Tetracyclines, Chloroquine, Thalidomide, Salicylates etc

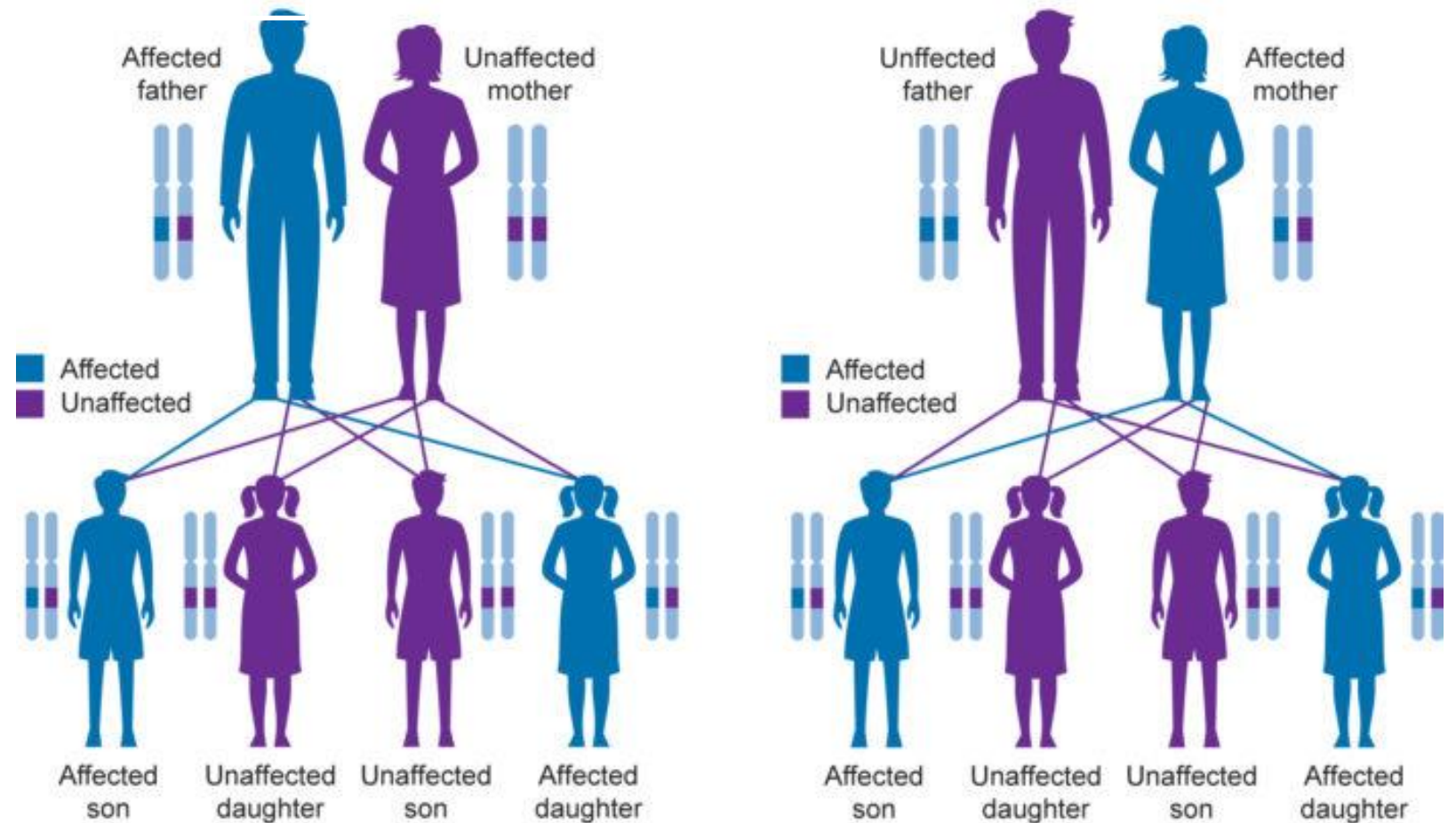
7. Maternal diseases & Defects, Uterine tumors, Uterine inflammation, Defects in implantation, Age, Emotional disturbances Stress Multiple pregnancies.

8. Abnormalities of the ovum, Abnormalities of semen
Antigen – antibody reactions



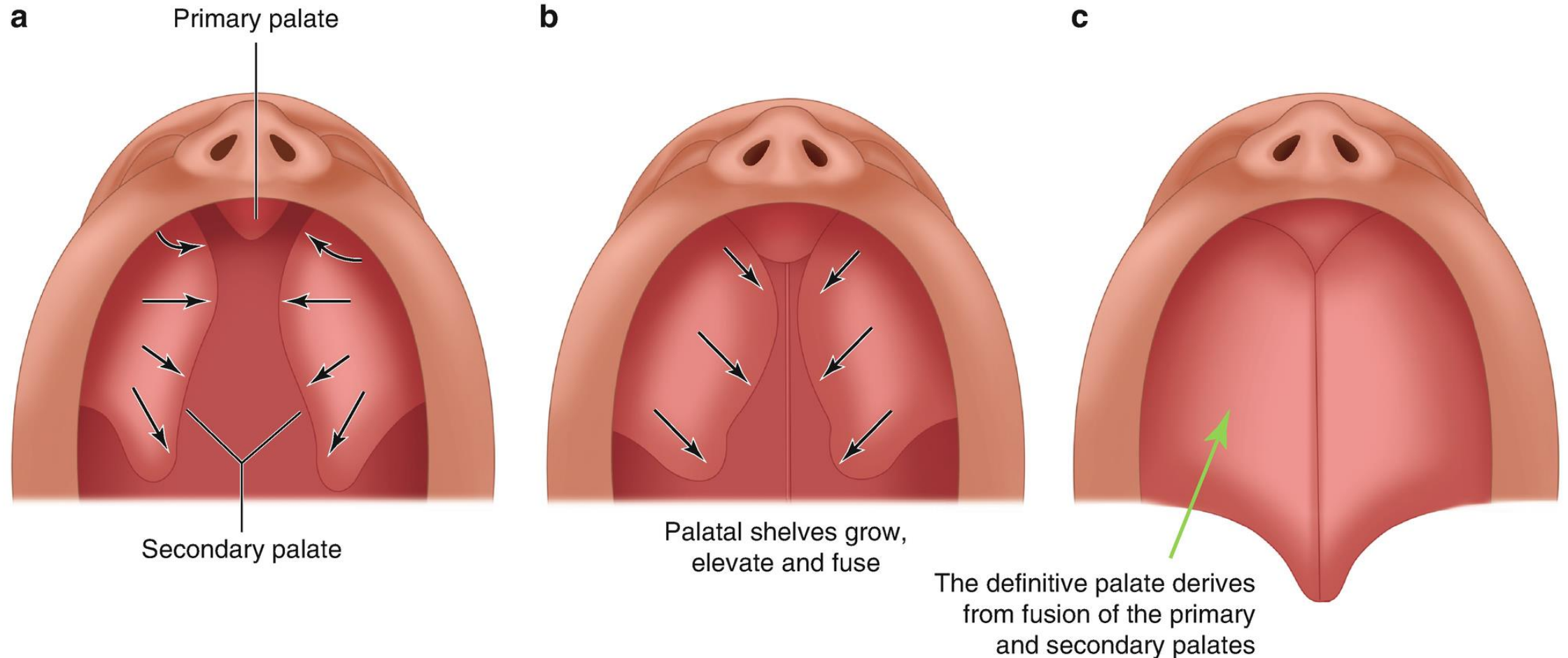
Causes of Developmental Disturbances

B. Genetic Inheritance Eg: Autosomal Dominant Inheritance



Example:

How Does the Developmental Disease Occur??



CLASSIFICATION OF DEVELOPMENTAL DISEASES

1. Based on time

- Prenatal – 1st or 2nd or 3rd trimester
- Post natal- Neonatal or Infancy or Toddler or Childhood or Adolescence.

2. Based on Etiology

- Intrinsic causes
- Extrinsic causes

3. Based on involvement of the oral and Paraoral tissues

Hard tissues

- Maxilla and mandible (Jaws)
- Teeth

Soft tissues

- Tongue
- Lips
- Cheeks
- Gingiva
- Salivary Glands

D/Disturbances of Hard Tissues

D/Disturbances of Maxilla and Mandible (Jaws)

1. Cleft palate
2. Micrognathia
3. Macrognathia
4. Hemifacial atrophy
5. Hemifacia Hypertrophy

D/Disturbances of Hard Tissues (Contd..)

D/Disturbances of Teeth

1. **Shape:** Fusion, Gemination, Concrescence, Dense invaginatus, Dens evaginatus, Dilaceration, Taurodontism, Enameloma, Talon cusp.
2. **Size:** Microdontia & Macrodontia.
3. **Number:** Anodontia, Supernumerary teeth,
4. **Structure:** Amelogenesis Imperfecta, Dentinogenesis Imperfecta, Dentin dysplasia.

D/DISTURBANCES OF SOFT TISSUES

Tongue

1. Macroglossia
2. Microglossia
3. Cleft-tongue
4. Ankyloglossia
5. Fissured tongue
6. Median rhomboid glossitis
7. Hairy tongue
8. Lingual thyroid nodule

Reference: [Electronic Link](#)

D/DISTURBANCES OF SOFT TISSUES (Contd..)

DEVELOPMENTAL DISORDERS OF LIPS

1. CLEFT LIP
2. LIP PITS
3. DOUBLE LIP

DEVELOPMENTAL DISORDERS OF CHEEKS

1. Fordyce granules
2. Leukoedema

DEVELOPMENTAL DISORDERS OF GINGIVA

1. RETROCUSPID PAPILLAE
2. GINGIVAL FIBROMATOSIS

DEVELOPMENTAL DISORDERS OF SALIVARY GLANDS

1. Aberrant Salivary Glands
2. Stafne's Cyst

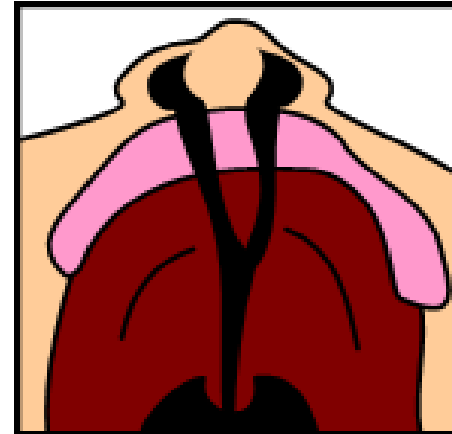
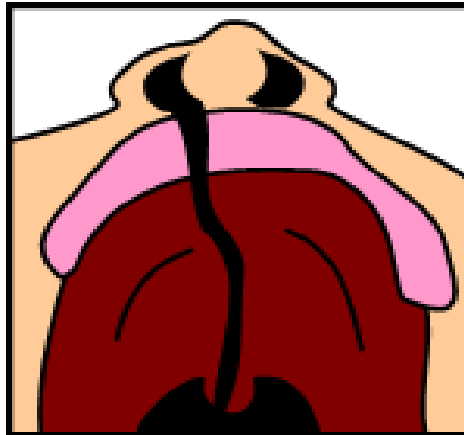
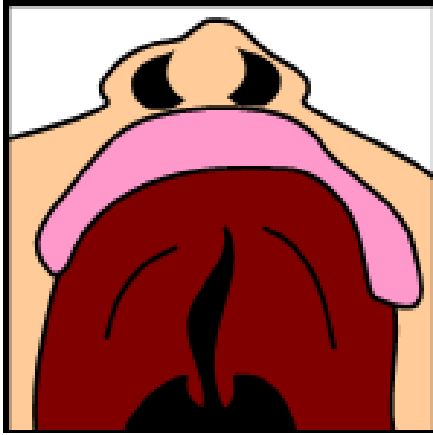
DEVELOPMENTAL DISEASES AFFECTING MAXILLA & MANDIBLE (JAWS)

CLEFT LIP and PALATE

- A condition in which a slit or a hole is present in the roof of the oral cavity.
- Cleft Palate occurs in about one in 700 live births worldwide.
- In most cases, cleft palate occurs along with cleft lip
- Either Syndrome associated or Non-Syndrome associated



Classifications



Davis and Ritchie classification.

Veau classification.

Kernahan and Stark symbolic classification.

International confederation of plastic and reconstructive surgery classification.



Etiopathogenesis

Multifactorial Threshold Hypothesis



Bixler Hypothesis:

when the total genetic liability of an individual reaches a certain minimum level, the threshold for expression is reached and a cleft occurs.

It is presumed that every individual carries some genetic liability for clefting, but if this is less than the threshold level, there is no cleft. When the individual liabilities of two parents are added together in their offspring, a cleft occurs if the threshold value is exceeded.

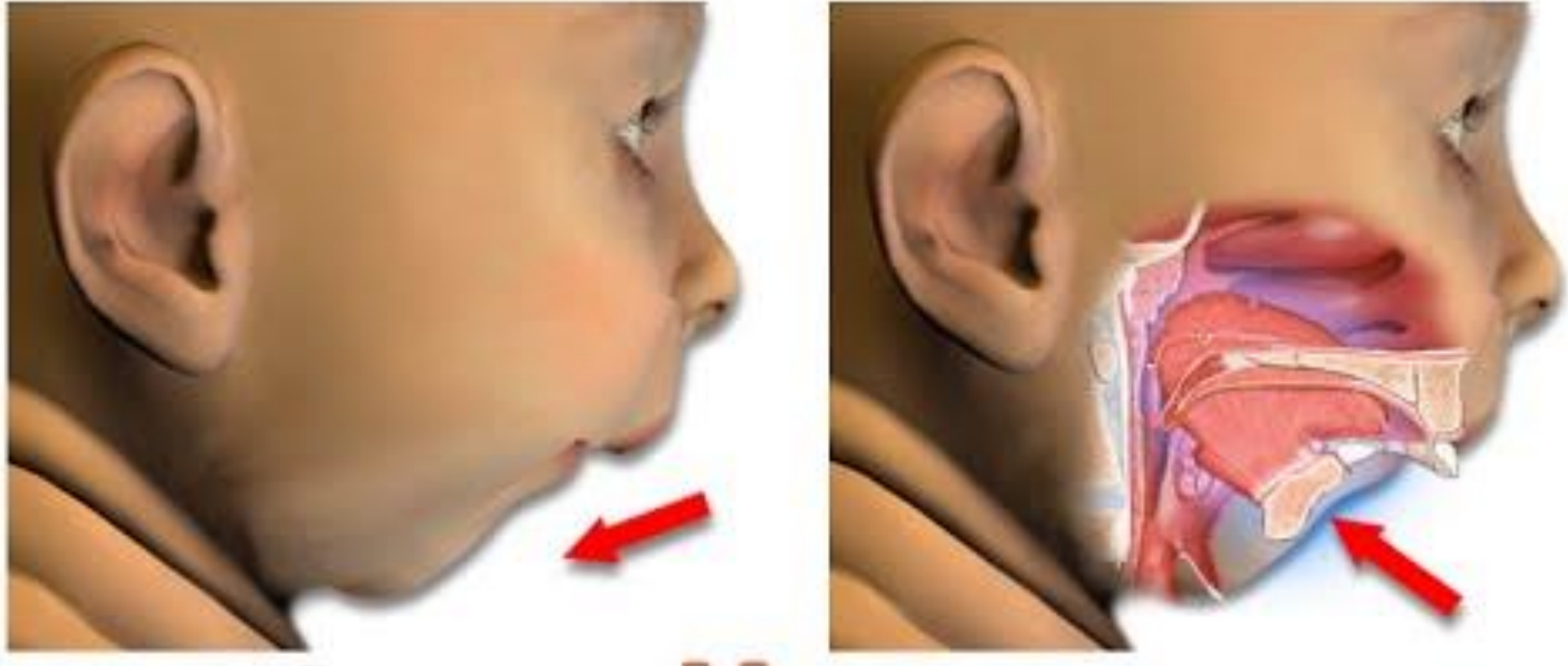
Complications

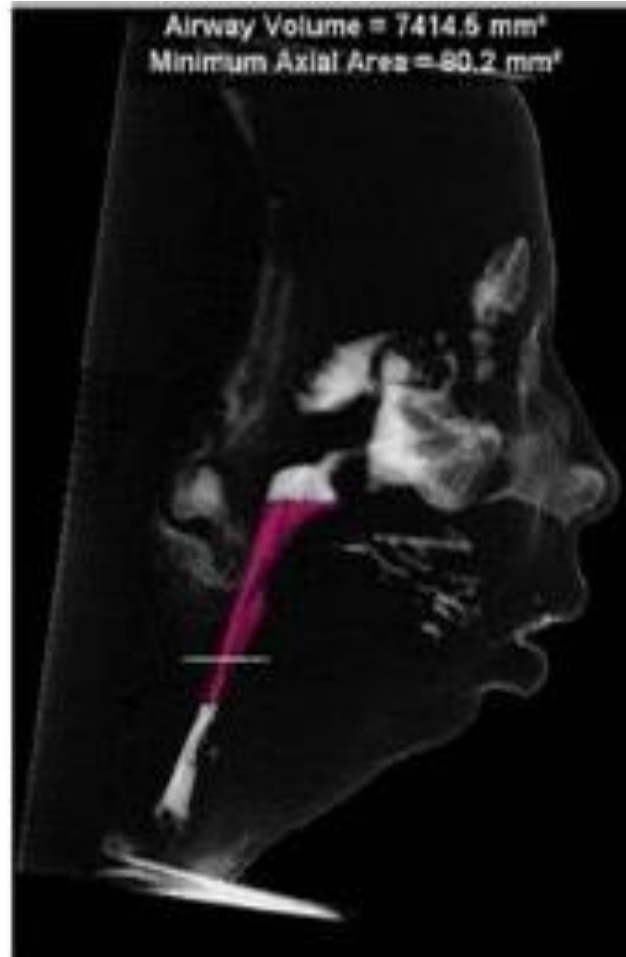


- Failure to thrive
- Sinus and Ear Infections
- Respiratory tract infections
- Abnormal Orofacial development

MICROGNATHIA

Congenital or Acquired (Traumatic)





- Pierre Robin Sequence
 - i. Small Jaw
 - ii. Tongue pushed into pharynx
 - iii. Airway Obstruction
- Oro-facial Development issues

HEMIFACIAL ATROPHY



Progressive unilateral trophy of face and underlying structures

Causes: Trauma, Dysfunction of peripheral nervous system, infection, and genetic abnormalities are suggested

Earliest sign: Cleft or furrow in midline of face

Parry Romberg Syndrome: Contralateral Jacksonian Epilepsy, trigeminal neuralgia, changes in eyes and hairs.



Other Developmental Diseases affecting Jaws

- Hemifacial Hyperthrophy
- Cherubism
- Osteopetrosis
- Osteogenesis Imperfecta
- Crouzon's Syndrome
- Cleidocranial Dysplasia
- Marfan Syndrome
- Treacher Collins Syndrome
- Ehlers-Donlos Syndrome
- Down Syndrome
- Fragile-X Syndrome



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DEVELOPMENTAL DISEASES OF TONGUE

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MACROGLOSSIA



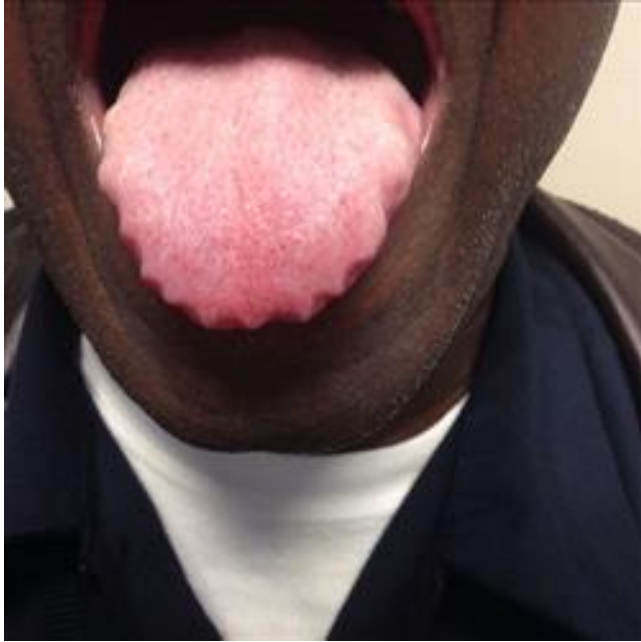
Macroglossia (enlarged tongue) is a condition that have tongues that are larger than typical, given the size of their mouths.

Macroglossia is linked to physiologic or to syndromes or to some forms of cancer or severe infections.

Common sign of macroglossia is sticking out tongue, Other signs and symptoms are:

- Noisy, high-pitched breathing (stridor).
- Snoring or low-pitched breathing (stertor).
- Difficulty eating or drinking
- Drooling of Saliva.
- Difficulty in speech.





Common Causes of Macroglossia:

1. Syndromes

- Down syndrome
- Hunter syndrome
- Beckwith-Wiedemann syndrome

2. Acquired Causes

- Amyloidosis
- Hypothyroidism
- Acromegaly
- Diphtheria

3. Neoplasms:

- Lymphangioma
- Hemangioma
- Lymphoma



Macroglossia treatment ranges from speech therapy to surgery.

MICROGLOSSIA



- Microglossia is a rare developmental condition that might impact the patient's respiratory, feeding and speech functions, in addition to other intraoral structures.
- Embryologically, the tongue has 2 origins (TI & HBE), which when affected, will determine whether the patient has microglossia or aglossia.
- The involvement of multiple factors, the presence of several anatomical anomalies and the growth exerted by patients, confer microglossia.
- A multidisciplinary team should adopt an organized approach in management.

ANKYLOGLOSSIA



Tongue-tie (ankyloglossia) is a condition present at birth that restricts the motion of tongue.

It is an unusually short, thick or tight lingual frenulum that tethers the bottom of the tongue to the floor of the mouth.

It may interfere with breastfeeding, difficulty sticking out the tongue and affect eating or speaking.

Some cases may require a surgical procedure for correction.



FISSURED TONGUE



A fissured tongue shows furrows or grooves on the surface of tongue.

Other names for a fissured tongue are lingua plicata, furrowed tongue or scrotal tongue.

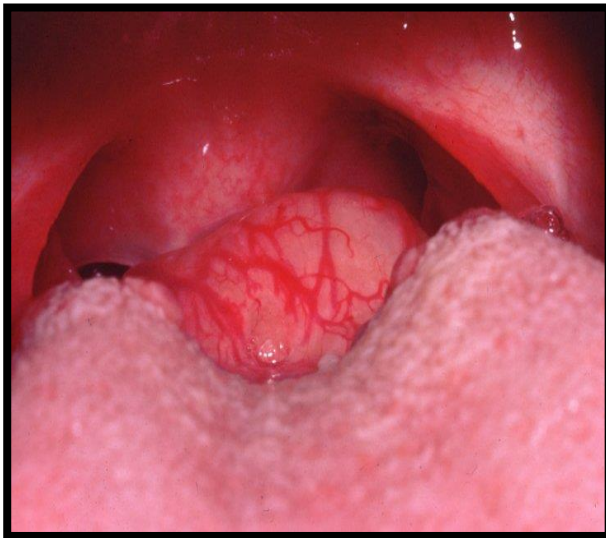
The grooves may be as shallow as 2 mm or as deep as 6 mm.

A fissured tongue cause symptoms due to entrapment of food or microorganisms:

- Halitosis.
- Glossitis.
- Discomfort or a burning Sensation.



LINGUAL THYROID NODULE



- Lingual thyroid is an abnormal mass of ectopic thyroid tissue seen in base of tongue caused due to embryological aberrancy in development of thyroid gland.
- It is caused due to failure in descendance of thyroid gland to its normal position during embryogenesis.
- About 70% of patients with ectopic thyroid are hypothyroid.
- It may manifest with symptoms such as dysphagia, dysphonia, and upper respiratory tract obstruction.

LINGUAL VARICOSITIES



- Lingual varicosities appear as tortuous, blue, red, and purple elevations that course over the ventrolateral surface of the tongue.
- They represent a degenerative change in the adventitia of the venous wall and are of no clinical consequence.
- They are painless and are not thought to be subject to rupture and/or haemorrhage.

HAIRY TONGUE



- Hairy tongue is abnormally long filiform papillae or keratin coating on the dorsal surface of the tongue.
- It is a common, temporary, and harmless condition that occurs in as much as 13% of the population.
- Bacteria and yeast can accumulate in the hair-like mesh. These accumulations result in various colours to the surface of the tongue.
- Hairy tongue may appear brown, white, green, or pink, depending upon the specific cause and other factors, such as mouthwashes or food colours.
- Certain types microorganisms can even give the tongue a black appearance, referred to as “black hairy tongue”

- **KINDLY REFER**

Shafer's text book of oral pathology – 2020 Elsevier (9th edition), Chapter – 2: Developmental Disturbances of Oral and Paraoral Structures Page: 10 to 71.

Oral Pathology - Clinicopathologic Correlation Regezi, Sciubba & Jordan;; 2017 Elsevier (7th edition), Chapter 16, Pg: 373-389

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DEVELOPMENTAL DISORDERS OF LIPS

- 1. CLEFT LIP**
- 2. LIP PITS**
- 3. DOUBLE LIP**

Cleft Lip



Surgical Treatment: follow a rule of 'three 10s' as requirement for identifying the child's status for surgery (i.e. 10 lb, 10 mg/L of hemoglobin, and age 10 weeks).

Cleft lip and palate are common congenital malformations.

1 in 500 to 1 in 2500 live births depending on geographic origin, racial and ethnic background and socioeconomic status.

Clefts can be Monogenic or Syndromic.

Etiology:

- Heredity
- Physiologic, emotional, or traumatic stress
- A defective vascular supply
- A mechanical disturbance due to size of the tongue
- Alcohol and certain drugs and toxins
- Infections
- Lack of inherent developmental force.

LIP PITS

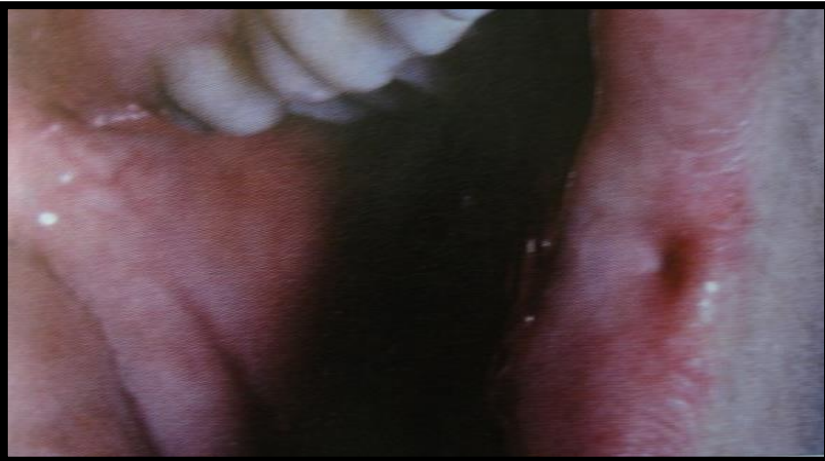
Para Median Lip Pits



Pits may result from notching of the lip at an early stage of development, with fixation of the tissue at the base of the notch, or from failure of complete union of the embryonic lateral sulci of the lip, which persist and ultimately develop into the typical pits.

Mucous secretion may exude from the base of this pit. The lip leading to swelling and accentuating the appearance of the pits.

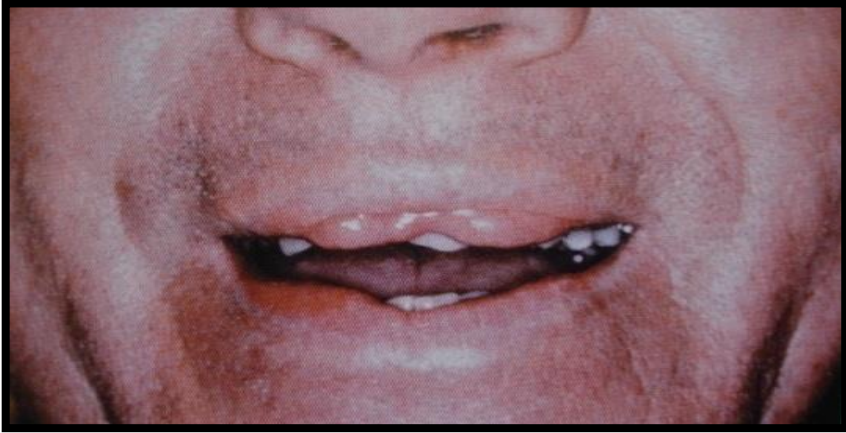
Commissural Lip Pits



The association of pits of the lower lip and cleft lip and/or cleft palate, termed van der Woude's syndrome.

Surgical excision of these various pits has been recommended but primarily for academic information, since the pits are harmless and seldom manifest complications.

DOUBLE LIP



Etiology

1. ***Congenital Causes***: Double lip can be a congenital anomaly often associated with other craniofacial syndromes, such as Ascher syndrome, which includes blepharochalasis (droopy eyelids) and thyroid gland abnormalities.
2. ***Acquired Causes***: Acquired cases may result from trauma, surgical scars, or irritation, which can lead to the formation of excess tissue.

Treatment

- ***Non-surgical Approaches***: observation may be sufficient, especially when the condition does not cause significant issues.
- ***Surgical Correction***: If the double lip is causing functional issues or significant aesthetic concerns.

DEVELOPMENTAL DISORDERS OF CHEEKS

1. Fordyce granules
2. Leukoedema

FORDYCE GRANULES



Developmental anomaly characterized by heterotopic collections of sebaceous glands at various sites in the oral cavity.

Fordyce's granules appear as small yellow spots, either discretely separated or forming relatively large plaques, often projecting slightly above the surface of the tissue.



They are found most frequently in a bilaterally symmetrical pattern on the mucosa of the cheeks opposite the molar teeth.

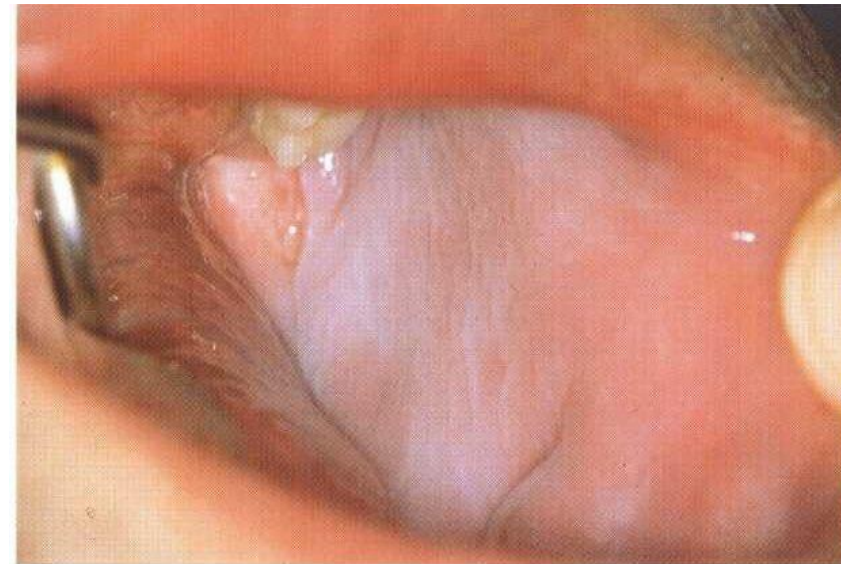
These glands are innocuous, have no clinical or functional significance and require no treatment.

LEUKOEDEMA



LEUKOEDEMA appears as filmy opalescence of the mucosa in the early stages to a more definite grayish-white cast with a coarsely wrinkled surface in the later stages.

Tobacco smoking was closely correlated with the occurrence of the leukoedema.



Microscopic features of leukoedema consist of an increase in thickness of the epithelium, intra-cellular edema, a superficial parakeratotic layer of several cells in thickness, and broad rete pegs.

Leukoedema appears to be simply a variant of normal mucosa, no treatment is necessary.

DEVELOPMENTAL DISORDERS OF GINGIVA

- 1. RETROCUSPID PAPILLAE**
- 2. GINGIVAL FIBROMATOSIS**

RETROCUSPID PAPILLAE



It is a small, elevated nodule located on the lingual mucosa of the mandibular cuspids.

It is exceedingly common in children, occurring in 99%

of those between the ages of eight and 16 years and appears to be regression of the structure with maturity.

It is a 'normal' anatomic structure which regresses with age and requires no-treatment.



GINGIVAL FIBROMATOSIS



Diffuse fibrous overgrowth of the gingival tissues, described for many years under a variety of terms.

In the majority of reported cases, the condition was hereditary, being transmitted through a dominant autosomal gene.

Clinically it appears as a dense, diffuse, smooth, or nodular overgrowth of the gingival tissues of one or both arches, usually appearing about the time of eruption of the permanent incisors.

Microscopic picture of the tissue in fibromatosis gingivae is similar to that of any fibrous hyperplasia.

Treatment: Surgical removal of the excessive tissue and exposure of the teeth are indicated

DEVELOPMENTAL DISORDERS OF SALIVARY GLANDS

- 1. Aberrant Salivary Glands**
- 2. Stafne's Cyst**

Stafne's Cyst

It is usually accidentally found via radiograph during other dental treatments because patients often present no abnormal clinical symptoms.

Etiology: It is caused by proliferation or translocation of adjacent structures such as salivary gland.

It is a benign, developmental bony defect causing no any pathological changes, surgical intervention is no longer needed to treat SBD



- **KINDLY REFER**

Shafer's text book of oral pathology – 2020 Elsevier (9th edition), Chapter – 2: Developmental Disturbances of Oral and Paraoral Structures Page: 10 to 71.

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